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1 Filling the Gaps: Helping Boilerexams Content Creators Identify Missing Video Explanations

1.1 Introduction and Problem

On the **Boilerexams** website, there are a number of exams that are missing video explanations. For those of you reading this who are spoiled by the video explanations in the Math courses on Boilerexams, go check out the CS251 exams. You might find questions with no videos explaining the reasoning behind the answers. As a CS major, I couldn't stand for this injustice, knowing I could do something about it.

Having studied using the CS251 material on Boilerexams, I knew the pain of trying to explain to friends why their answer was correct or wrong. Without video explanations, we had to piece things together ourselves. That's when I decided to build a tool that would **identify which questions were missing videos**, so the content team could quickly snipe those questions and create videos for them.

1.1.1 Methodology

The content team is made up of Purdue students; as a result, they only have so much time. Therefore, if I wanted to both help students and make good use of the content team's time, I needed to **identify the most optimal questions**.

I came up with three metrics to prioritize which questions needed videos: 1. Average Percent Correct – How often students got the question right. 2. Total Submissions – How many times the question was attempted. 3. Number of Correct Submissions – How many of those attempts were correct.

1.1.2 Building the Table

To make this work, I needed to gather data from multiple tables in the database. I combined the Questions, Exam, Course, and Video tables to build a complete list of questions without videos.

Here's where it got tricky: some questions were associated with multiple topics. For example, a shell sorting question might be under both Sorting and Shell Sorting. To clean this up, I grouped all the duplicate questions together so that each question appeared only once, but with all its associated topics listed.

Insert screenshot of table

With the table ready, I moved on to making the tool useful for the content team.

1.1.3 Building the Interface

A table format was okay, but not simply put not good enough. I wanted the tool to be even easier for the content team to use. Therefore, I built an interface using Jupyter Notebook's ipywidgets. The interface lets the team:

- Select a course they're interested in.
- Sort questions by any of the three metrics to focus on the most important questions.

Insert screenshot of final Jupyter IPywidget

This interface also allows the content team to click a link to a question to get context on the question.

1.2 What's Next?

This tool was a great start. There are some further steps that can be taken to improve the product. Here are a few ideas: 1. Integration with the Course Creation Platform – Embedding the tool directly into the course creation page so the content team doesn't have to load up the notebook. 2. Real-Time Insights on Most-Missed Questions – Adding a feature that tracks a running average of the most-missed questions leading up to an exam. This would be helpful for: - For courses with fewer exams or that are newer, the tool helps prioritize missing videos for upcoming exams based on recent student struggles. - For courses with many exams, it focuses on aligning video creation with current student needs by analyzing recent data.

1.3 Why This Matters?

This tool is something that can **augment the content team's time and effort** by giving them the data they need to make smart decisions about where to focus their work. More importantly, this tool aims to reduce the number of questions missing video explanations — something I wish I had when I was taking CS251.